

Amrita Vishwa Vidyapeetham
Amrita School of Computing
Course Plan and Evaluation Plan

Course code	23ECE205
Course name	DIGITAL ELECTRONICS
Academic year	2025 – 26
Program	B.Tech CSE
Semester	Third (Odd)
Year of admission & Batch	2024
No. of students	A(63) ,B(60), C(62)
Faculty Advisors	
Faculty	Ms Sangita Khare (A & B) & Dr.Kavya Sai(C)
L T P CR	3 0 0 3
Total number of hours planned	50 hours

Course Objectives

- To provide an understanding of basic building blocks of digital circuits
- To enable the understanding of Boolean algebra and logic function optimization
- To enable design of combinational and sequential circuits

Course Outcomes

- CO1: Realize a given expression in terms of basic building blocks.
- CO2: Minimize a given logic expression.
- CO3: Design combinational circuits
- CO4: Design Sequential circuits.

CO-PO Mapping

PO/ PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO13	2												2	
CO23	2												2	
CO33	3	2										2	2	
CO43	3	2										2	2	

Syllabus

Unit 1: Introduction to logic circuits - Variables and functions, inversion - Truth tables - Logic gates and Networks - Boolean algebra - Synthesis using gates - Design examples - Optimized implementation of logic functions - Karnaugh map - Strategy for minimization - Minimization of product of sums forms - Incompletely specified functions - Multiple output circuits - Tabular method for minimization.

Unit 2: Combinational circuit building blocks - Number representation and arithmetic circuits: Addition of unsigned numbers - Signed numbers - Fast adders - Multiplexers - Decoders - Encoders - Code converters - Arithmetic comparison circuits.

Unit 3: Sequential circuit building blocks - Basic latch - Gated SR latch - Gated D latch - Master slave and edge triggered - D flip-flops - T flip-flop - JK flip-flop, Registers, Asynchronous Counters, Synchronous Counters, Ring Counter and Johnson Counter, Synchronous sequential circuits - Basic design steps - State assignment problem – Design of Mealy and Moore state models.

Textbook(s)

Stephen Brown, Zvonko Vranesic, "Fundamentals of Digital logic with Verilog Design", Tata McGraw Hill Publishing Company Limited, Special Indian Edition, 2007.

R. D. Sudhakar Samuel, "Logic Design: A Simplified Approach", Sangvine Technical Publishers, Edition 1, 2006

Reference(s)

M Morris Mano and Michael D Ciletti, "Digital Design with Introduction to the Verilog HDL", Pearson Education, Fifth Edition, Fifth Edition, 2015

John F. Wakerly, "Digital Design Principles and Practices", Fourth Edition, Pearson Education, 3rd Ed, 2008.

Donald D Givone, "Digital Principles and Design", Tata McGraw Hill Publishing Company Limited, 2003.

Course Plan - Detailed Lecture Plan:

Week. No	Working		Topic	Evaluation/ Activities
	Day No.	Date period		
1	1-2	July	Introduction to logic circuits - Variables and functions, inversion - Truth tables	
2	3-5	July	- Logic gates and Networks -	

Week. No	Working Day No.	Date period	Topic	Evaluation/ Activities
3	6-8	July	Boolean algebra - Synthesis using gates - Design examples	
4	9-14	August	- Optimized implementation of logic functions - Karnaugh map - Strategy for minimization -	Quiz 1
5	15-18	August	Minimization of product of sums forms - Incompletely specified functions - Multiple output circuits - Tabular method for minimization.	
6	19-20	August	Combinational circuit building blocks	
7	21-24	August	Number representation and arithmetic circuits: Addition of unsigned numbers - Signed numbers- Fast adders	
8	25-30	September	Multiplexers - Decoders - Encoders	
9	31-32	September	- Code converters - Arithmetic comparison circuits.	
10	33-35	September	Sequential circuit building blocks - Basic latch - Gated SR latch - Gated D latch -	
11	36-38	September	Master slave and edge triggered -	
12	39-42	October	D flip-flops - T flip-flop - JK flip-flop, Registers,	Quiz 2
13	43-44	October	Asynchronous Counters, Synchronous Counters,	
14	45	October	Ring Counter and Johnson Counter,	
15	46-47	October	Synchronous sequential circuits -Basic design steps - State assignment problem.	Missed Mid-Sem Exam
16	48-49	November	– Design of Mealy and Moore state models	Quiz 3, Pre-registration, Course End Survey, and Faculty Feedback

Week. No	Working Day No.	Date period	Topic	Evaluation/Activities
17	50	November	Revision	Last Instruction Week
18	-		-	End-Sem Exam

Evaluation Pattern: 60:40

Assessment	Internal	External
Mid Sem Examination	30	--
*Continuous Assessment (CAT)	30	--
End Semester	--	40

**CAT includes Quizzes, Assignments and Tutorials*

Evaluation Plan

Component	Number of Evaluations	Weightage
CAT (Internal)	03	30%
Mid-Term Exam (Internal)	01	30%
End Term Exam (External)	01	40%

COs to Evaluations Mapping

Evaluation		Max marks	Weightage	CO Mapping	Tentative schedule
CAT	Quiz 1	10	10	CO1	Last week of july
	Quiz 2	10	10	CO3	Last week of august
	Quiz 3	10	10	CO4	3 rd week of october
Mid term		50	30	CO1-CO3	
End term		100	40	CO1-CO4	

CO	CAT (30%)			Mid-Term	End-Term	Mark
	Q1	Q2	Q3	(30%)	(40%)	distribu-
Marks	(10%)	(10%)	(10%)			tion (in
	10	10	10	50	100	%)
CO1	6 (10)	-	-	10 (06)	10 (04)	16
CO2	4	6	-	10 (06)	10 (04)	20
CO3	-	4 (10)	-	30 (18)	30 (12)	34
CO4	-	-	10 (10)	-	50 (20)	30

(Ms. Sangita Khare & Dr. Kavya Sai)

Name and Signature of Faculty Name and Signature of Chairperson